Run report: automorphism rho on the round-2 crater curves

Run start (UTC). 2026-10-15T23:05Z. **Report type.** Descriptive run report (observations only). Interpretation is in the study README.

What ran

Protocol	`research/prime_rho_automorphism_20261015/PROTOCOL.md`, frozen before the run
Specs	`specs/cubic.json` (`ECS1haae1b01f9883`), `specs/quartic.json` (`ECS1h26c7bf2d77bd`)
Sessions	`ECBS1h5e1b9912a46d` (cubic), `ECBS1hf8d1171784f1` (quartic)
Comparisons	`ECC1hb42c42de3b9f` (cubic), `ECC1hcc414017408e` (quartic)
Source	commit `c9db37d01bc776a1e8594ce5495e58ba2ccf5a1d`, `git_dirty: false`
Binary	`ecbench` SHA-256 `0a8a48d7…1473b6`
Host	AMD EPYC 9R05 VM, 2 logical cores, 1 NUMA node, Linux 6.18 x86_64; env class `ECBENV2h0e1f15f7c61e`
Arms	`rho.negation` (reference, A = 2) vs `rho.cubic` (A = 6) on six j = 0 curves; `rho.negation` vs `rho.quartic` (A = 4) on six j = 1728 curves
Workloads	the twelve crater curves of round-2 E2, 4 targets each, target seed 20261011
Measurement	3 alternating rounds, 1 warm-up, seed 1, 900 s timeout, L2 isolation requested

Completion

Each session planned and wrote 192 records, and all 192 are `verified`. There were no failures and no timeouts. `ecbench verify --replay 12` reproduced 12 of 12 replays with 0 problems for each session.

Isolation earned: L0 on all 144 paired attempts (L2 was requested). The harness recorded a p95 foreign busy share of 82.5% (cubic) and 83.0% (quartic). Wall-time comparisons are therefore **descriptive, not results**, because 0 pairs were admitted and at least 5 are needed. The harness's measurement quality score is 0 / 100 for both.

Measured values

The unit is `ecbench.gae` (S, group-addition equivalents). Values are ratios of mean S, candidate over reference. Intervals are 95% cluster-bootstrap intervals with 4,000 resamples, 72 same-seed pairs, same session.

comparison	ratio (measured)	95% interval	predicted (derived)
`rho.cubic` / `rho.negation`	0.6112	[0.505, 0.739]	$1/\sqrt{3} = 0.577$
`rho.quartic` / `rho.negation`	0.7910	[0.673, 0.923]	$1/\sqrt{2} = 0.707$

The predicted values are derived from the expected-steps model $\sqrt{(\pi r / 2A)}$ with A = 6 or 4 against A = 2, as frozen in the protocol. The harness labels both ops ratios **"diagnostic only (unpriced work; ratio does not bound full cost)"**. The cost of mapping each point to its class representative is not priced in S.

Wall median ratios (descriptive only, not admitted): 0.980 (cubic) and 0.990 (quartic).

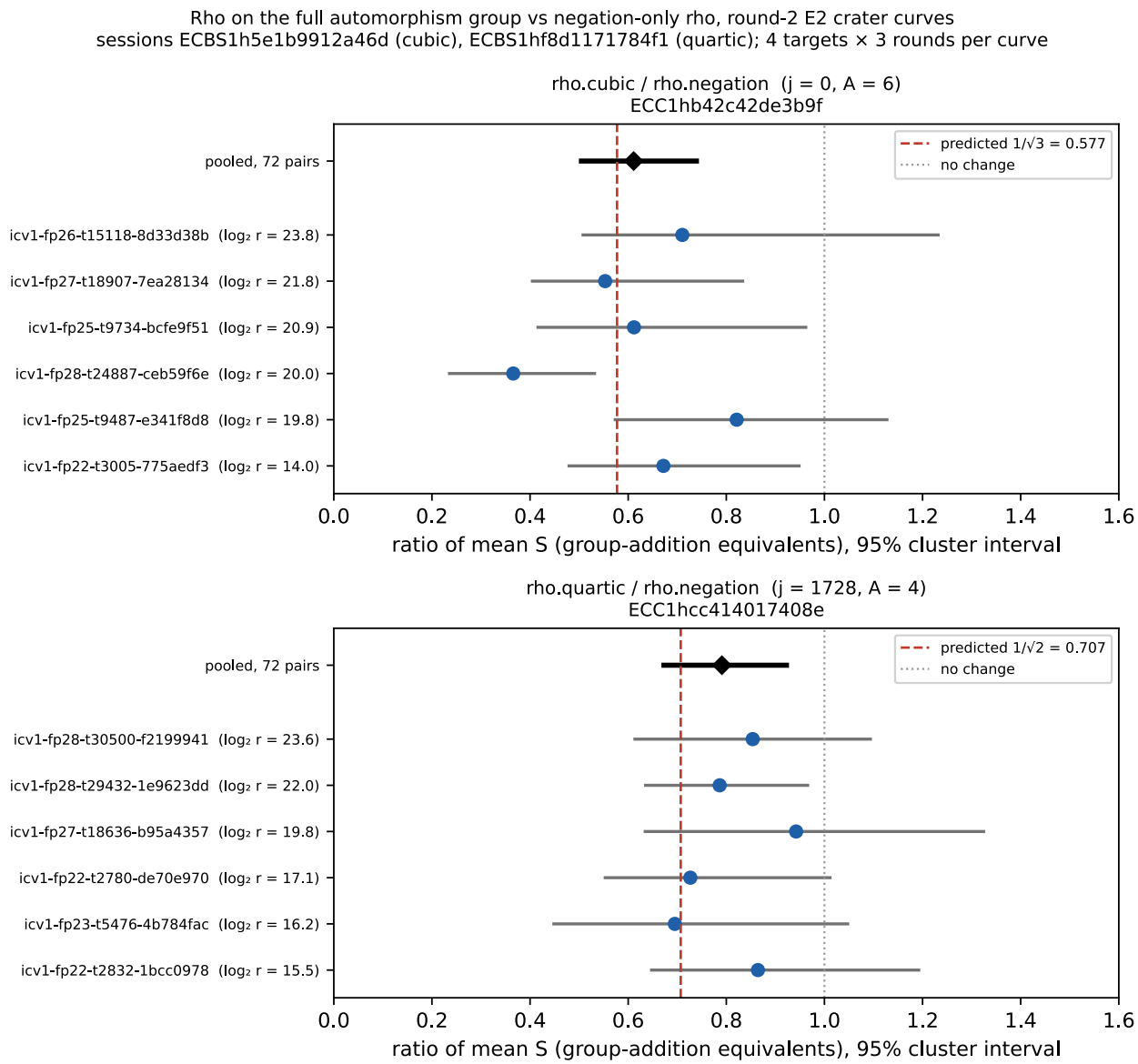


Figure 1: Per-curve and pooled ratios of mean S with 95% cluster intervals. Each curve has 12 pairs. Comparisons ECC1hb42c42de3b9f and ECC1hcc414017408e. Plotted from the saved comparison JSON. No values were recomputed.

Raw records

- `research/prime_rho_automorphism_20261015/sessions/{cubic,quartic}/records.jsonl`
- `…/sessions/{cubic,quartic}/comparisons/*.json`
- `…/sessions/{cubic,quartic}.audit.json` and `.verify.log`
- `…/sessions/{cubic,quartic}.table.md`

Missing data

- No L2-admitted wall-time interval exists. The host was busy with foreign load.
- There is no full-cost (priced) comparison. `ecbench price-units` covers Koblitz curves only.